



Points:
1/1

1. An education app calculates the grade based on marks:

```
marks = 82

if marks >= 90:
    grade = 'A'
elif marks >= 80:
    grade = 'B'
elif marks >= 70:
    grade = 'C'
else:
    grade = 'F'

print(grade)
```

What will be the student's grade?



Points:
4/4

2. Ritu is building an inventory tracker for a stationery shop. Each item is stored as:

```
item = {
    "name": "Marker",
    "quantity": 12,
    "price": 15.5,
    "in_stock": True
}
```

Which of the following are valid data types used in `item` ?

(Select all that apply)

- ☒ `bool` ✓
- ☒ `float` ✓
- ☒ `int` ✓
- ☒ `str` ✓



Points:
1/1

3. A fitness app checks if a person is a premium member **and** has an active subscription:

```
user_type = "Premium"
active = True

if user_type == "Premium":
    if active:
        print("Access granted")
```

Which combinations will still grant access?

- ☒ `user_type == "Premium" and active == True` ✓
- ☐ `user_type == "Basic" and active == True`
- ☐ `user_type == "Premium" and active == False`
- ☐ `user_type == "Guest"`



Points:
4/4

4. Nisha is building a budgeting app and logs each item in a dictionary.

```
entry = {  
    "item": "Pen",  
    "quantity": 2,  
    "cost": 12.5,  
    "in_stock": True  
}
```

Match the correct pairs:

True

Boolean



12.5

Float



2

Integer



Pen

String



Points:
1/1

5. Sanya is building a shopping cart. She wants to remove all instances of the product "pen" from the cart list:

```
cart = ["pen", "notebook", "pen", "pencil"]  
while "pen" in cart:  
    cart.remove("pen")  
print(cart)
```

What will be printed?

- ☐ ["notebook", "pen", "pencil"]
- ☐ ["pencil", "notebook"]
- ☐ ["pen", "pen", "notebook", "pencil"]
- ☒ ["notebook", "pencil"] ✓



Points:
3/3

6. An invoice system logs time like this:

```
from datetime import datetime  
now = datetime.now()  
print(now.strftime("%l:%M %p"))
```

What do the format codes mean?

(Select all the three right answers)

- ☐ %H → Minutes
- ☒ %p → AM/PM ✓
- ☒ %I → 12-hour format ✓
- ☒ %M → Minutes ✓

✗

Points:
0/1

7. Manoj wants to read a file that may or may not exist:

try:

```
with open("data.csv", "r") as file:  
    content = file.read()
```

```
except FileNotFoundError:  
    print("File not found.")
```

If `data.csv` is missing, what will be printed?

- ☐ Nothing is printed
- ☐ The program crashes
- ☒ **File not found. ✗**
- ☐ Error

✗

Points:
0/1

8. Priya is developing a smart home control system. Two devices respond to these binary signals:

```
light = 10 # 1010  
fan = 4    # 0100  
signal = light ^ fan  
print(signal)
```

What will be the final control signal output?

8 ✗

Correct answer: 14

✗

Points:
0/1

9. A math app uses a nested loop to generate multiplication tables for numbers 1 to 3:

```
for i in range(1, 4):  
    for j in range(1, 4):  
        print(i * j, end=" ")  
    print()
```

How many multiplication values will be printed?

3 ✗

Correct answer: 9

✓

Points:
1/1

10. A quiz app selects a random question from a list:

```
import random  
  
questions = ["Q1", "Q2", "Q3", "Q4"]  
print(random.choice(questions))
```

What will this print?

- ☒ **Any one from "Q1", "Q2", "Q3", "Q4" ✓**
- ☐ Always "Q1"
- ☐ Index of the selected question
- ☐ Error due to wrong syntax



Points:
1/1

11. Tina wants to maintain a transaction log. Which mode will help her **add new entries without overwriting** old ones?

```
with open("log.txt", "____") as f:  
    f.write("Transaction complete\n")
```

- ☐ "r"
- ☒ **"a"** ✓
- ☐ "w"
- ☐ "x"



Points:
1/1

12. Aman is collecting survey data and writing to a file. To avoid crashes on input error, he uses:

```
try:  
    age = int(input("Enter age: "))  
    with open("survey.txt", "a") as f:  
        f.write(f"Age: {age}\n")  
except ValueError:  
    print("Invalid input! Please enter a number.")
```

What type of error is this code trying to prevent?

- ☒ **ValueError** ✓
- ☐ NameError
- ☐ TypeError
- ☐ FileNotFoundError



Points:
3/3

13. An HR tool converts a joining date to a readable format:

```
from datetime import datetime  
join_date = datetime(2025, 7, 1)  
print(join_date.strftime("%d/%m/%Y"))
```

Which format specifiers are used correctly?

(Select all three right answers)

- ☒ **%d → Day** ✓
- ☒ **%m → Month** ✓
- ☐ %M → Month
- ☒ **%Y → Year** ✓



Points:
1/1

14. While displaying password hints, Tina wants to show the reverse of the user's password without revealing it directly.

```
password = "Secret123"  
hint = password[::-1]  
print(hint)
```

What will be printed?

- ☒ **321terceS** ✓
- ☐ None
- ☐ Error
- ☐ Secret123



Points:
1/1

15. A supermarket manager uses a dictionary to track product quantities:

```
inventory = {'apples': 20, 'bananas': 35, 'oranges': 15}
inventory['bananas'] += 5
print(inventory['bananas'])
```

What will be the output?

- ☒ 40 ✓
- ☐ 45
- ☐ 5
- ☐ 35



Points:
1/1

16. A travel agency gives different fares based on age and class:

```
age = 65
travel_class = "Sleeper"

if age >= 60:
    if travel_class == "Sleeper":
        fare = 100
    else:
        fare = 150
else:
    fare = 200

print(fare)
```

What fare will be printed?

100 ✓



Points:
1/1

17. An automation script checks if a folder exists before proceeding:

```
import os

if os.path.exists("myfolder"):
    print("Folder found.")
else:
    print("Folder missing.")
```

If the folder **doesn't exist**, what will be printed?

- ☐ FileNotFoundError
- ☐ None
- ☐ Folder found.
- ☒ Folder missing. ✓



Points:
0/1

18. A tuple is used to store seat details in the format:

```
(seat_number, passenger_name, seat_class)

seat = (23, "Rahul", "AC")
number, name, _ = seat
print(name)
```

What will be the output?

- ☒ seat ✗
- ☐ "Rahul" ✓
- ☐ "AC"
- ☐ 23



Points:
1/1

19. A game developer wants to create a countdown timer using a `while` loop:

```
count = 3
while count > 0:
    print(count)
    count -= 1
```

What is the last number printed before the loop stops?

1



Points:
1/1

20. Anup is building a logout system for a web app. He wants to reset the user's access token using:

```
token = 17 # or any number
reset_token = token ^ token
print(reset_token)
```

What value will be assigned to `reset_token` after the operation?

0



Points:
1/1

21. The bank ATM system verifies balance before allowing a transaction:

```
balance = 5000
withdraw = 4500

if balance >= withdraw:
    if withdraw % 100 == 0:
        print("Transaction Approved")
    else:
        print("Amount must be in multiples of 100")
else:
    print("Insufficient Balance")
```

What will be printed?

Transaction Approved



Points:
1/1

22. Ravi is building a messaging app where certain offensive words must be replaced with `***`. He uses the following logic:

```
msg = "This is a dumb idea"
clean_msg = msg.replace("dumb", "****")
print(clean_msg)
```

What is the output?

- ☐ This is idea
- ☐ This is a ***
- ☒ This is a **** idea ✓
- ☐ This is a dumb idea



Points:
1/1

23. You compare two different data structures:

```
a = (1, 2)
b = [1, 2]
print(a == tuple(b), a == b)
```

What is the output?

- ☐ False False
- ☐ False True
- ☒ True False ✓
- ☐ True True



Points:
1/1

24. A security system uses a bitmask to check access permissions. Raj is debugging the following code:

```
admin = 6    # binary: 110
guest = 3    # binary: 011
print(admin & guest)
```

What is the result of this bitwise AND operation?

2 ✓



Points:
1/1

25. You run the following Python code:

```
x = 5
y = 5.0
print(x == y, x is y)
```

What is the output?

- ☒ True False ✓
- ☐ False True
- ☐ False False
- ☐ True True



Points:
1/1

26. A developer writes the following code to check login status:

```
username = "admin"
password = "admin123"

if username == "admin":
    if password == "admin123":
        print("Login Successful")
    else:
        print("Invalid Password")
else:
    print("Invalid Username")
```

What is the output for the given values?

Login Successful ✓



Points:
1/1

27. An elevator panel encodes floor levels using a number. To reach a certain level, the control panel shifts the bit to the right:

```
floor = 8 # binary: 1000
print(floor >> 2)
```

Which floor number will it now represent?

2 ✓



Points:
1/1

28. Neha wants to generate a random 4-digit OTP using the `random` module:

```
import random
otp = random.randint(1000, 9999)
print(otp)
```

Which of the following is **true**?

- ☐ This code will raise an error
- ☐ The OTP will always be 9999
- ☐ OTP starts at 0
- ☒ **OTP can be any number between 1000 and 9999 (inclusive) ✓**



Points:
1/1

29. Vikram is creating a school timetable using a nested list:

```
timetable = [
    ["Math", "English", "Science"],
    ["History", "Math", "PE"]
]
print(timetable[1][2])
```

Which subject is scheduled in the 3rd period of Day 2?

- ☐ Science
- ☒ **PE ✓**
- ☐ History
- ☐ Math



Points:
1/1

30. An administrator stores student roll numbers as a tuple to ensure they aren't accidentally modified:

```
roll_numbers = (101, 102, 103, 104)
roll_numbers[2] = 110
```

What will happen when this code runs?

- ☐ It changes the third roll number to 110
- ☐ It removes the third roll number
- ☒ **It raises a TypeError ✓**
- ☐ It executes without any issue



Points:
1/1

31. Manoj is building an app that shows how many days are left until a user's birthday:

```
from datetime import date
today = date(2025, 6, 21)
birthday = date(2025, 12, 25)
delta = birthday - today
print(delta.days)
```

What will be printed?

- ☐ Error
- ☐ 180
- ☒ **186 ✓**
- ☐ 187

✗

Points:
0/1

32. You are writing a Python script that stores various types of data in a list.

```
data = [10, "10", 10.0, True]
```

Which of the following statements is **True**?

☒ **data.count("10")**

it will returns 2 ✗

☐ "10" + 10

it will execute without error

☐ **data.count(10)**

it will returns 3 ✓

☐ 10 == True

it will evaluates to True

✗

Points:
0/2

33. Neha's program reads a report file and catches errors if the file is missing:

```
try:  
    f = open("report.txt")  
    print(f.read())  
    f.close()  
except FileNotFoundError:  
    print("File is missing")
```

Which are true about this code? (Select both the right answers)

☐ It avoids program crash if the file is missing ✓

☒ **It uses with open block ✗**

☒ **It uses try-except for safe access ✓**

☐ It writes data into the file

✓

Points:
1/1

34. A school arranges 2 rows with 3 seats each. The layout is printed using nested **for** loops:

```
for row in range(1, 3):  
    for seat in range(1, 4):  
        print(f"Row {row} Seat {seat}")
```

What will be the total number of seat labels printed?

✓

✓

Points:
2/2

35. A teacher logs roll numbers of students using a loop:

```
for i in range(1, 6):  
    print(f"Student Roll No: {i}")
```

Which statements are true?

(Select both the right answers)

☐ It runs infinitely

☐ It prints from 0 to 5

☒ **The loop executes 5 times ✓**

☒ **It prints roll numbers from 1 to 5 ✓**



Points:
1/1

36. Rashmi represents a small movie theater layout using a nested list, where **1** means the seat is booked:

```
seats = [  
    [1, 0, 0],  
    [0, 1, 1],  
    [0, 0, 0]  
]  
print(sum(row.count(1) for row in seats))
```

How many seats are booked in total?

- ☐ 3
- ☐ 2
- ☒ 4 ✓
- ☐ 5



Points:
1/1

37. An event planner uses a dictionary where each **event date (day, month)** is the key, and the value is the event name:

```
events = {  
    (15, 'Aug'): 'Independence Day',  
    (26, 'Jan'): 'Republic Day'  
}  
print(events[(15, 'Aug')])
```

What will be printed?

- ☐ 15, Aug
- ☐ Republic Day
- ☒ Independence Day ✓
- ☐ KeyError



Points:
1/1

38. Ritu is organizing a 3-day workshop and is storing participant names for each day in a single list:

```
participants = ["Ankit", "Sneha", "Ravi", "Neha"]  
participants.insert(2, "Priya")  
print(participants)
```

What will be the output?

- ☐ ["Priya", "Ankit", "Sneha", "Ravi", "Neha"]
- ☒ ["Ankit", "Sneha", "Priya", "Ravi", "Neha"] ✓
- ☐ ["Ankit", "Sneha", "Ravi", "Neha", "Priya"]
- ☐ ["Ankit", "Sneha", "Ravi", "Priya", "Neha"]



Points:
2/4

39. A school reporting system uses formatted dates for attendance reports.

21/06/2025 ✓

03:45 PM ✓

Saturday ✗

Correct answer: %A

Jun ✗

Correct answer: %b



Points:
4/4

40. In an online student form, Neha encounters various errors while validating inputs and file access.

Match the correct pairs:

Name input expected but integer entered

TypeError



File not present

FileNotFoundError



int() used on a string like "abc"

ValueError



Dividing total by zero in average calc

ZeroDivisionError

